

Team Members: Carlos Mejia, Ashley Prado, Gabriel Garcia, Marcelo Hernandez

Project Overview:

Sequel is a prototype warehouse inventory management system designed to simulate the backend operations of a commerce platform. The system tracks inventory items, warehouses, suppliers, employees, and transactions using a relational PostgreSQL database.

The project is organized into 3 layers: a database layer containing the schema, sample data, and SQL queries; a Python backend that connects to the database using the psycopg2 library; and an HTML-based user interface for interacting with the system.

The goal of Sequel is to bring structure to warehouse operations by allowing inventory data to be stored, queried, and analyzed through well defined database relationships. Users can manage product inventory, monitor storage locations, and track sales and movement activity within a warehouse environment.

Schema Summary:

The database consists of 11 tables. The core entities are *Person*, *Employee*, *Warehouse*, *Address*, *InventoryItem*, *StorageLocation*, *Supplier*, *PurchaseOrder*, *Sale*, *SaleItem*, and *Refund*.

Person serves as the base entity for both customers and employees. *Employee* extends *Person* via a one-to-one relationship and is assigned to a *Warehouse*. Each *Warehouse* has a physical *Address* and contains *InventoryItem* records organized by *StorageLocation*. Suppliers provide inventory and are linked through *PurchaseOrder* to a specific warehouse. On the sales side, a *Person* (as customer) has many Sales, each broken into line items via *SaleItem* (linking to *InventoryItem*), and optionally reversed through a *Refund*.

Advanced Features (Marcelo):

- Database view to get sales and their details (simplified Ashley's query):
<https://github.com/redhatgamer/Sequel/commit/e92282324be44aa17aad9d22c2963ef854a9adba>
- Database transaction, to make a purchase in the inventory tab:
<https://github.com/redhatgamer/Sequel/commit/d5357638ea116466d24066eb5c74a785d4acafa7>

Team Members: Carlos Mejia, Ashley Prado, Gabriel Garcia, Marcelo Hernandez

Individual Contribution Statement:

	Carlos Mejia	Ashley Prado	Gabriel Garcia	Marcelo Hernandez
Tables Worked On	Worked on <i>PurchaseOrder & Supplier.</i>	InventoryItem, Sale, SaleItem, Refund.	Design phase for ERD. Constraints testing.	
Queries Written	Flask API to React Frontend Connection Real-Time Search and Filtering Dark-Themed Responsive UI Design	Inventory report, low-stock query, sales history, sales item details, refund report, supplier inventory summary.	Complete Order Fulfillment View Inventory Value by Warehouse Sales by Customer with Totals	Purchase order status report. PurchaseOrder sample data
Backend Features Implemented	React.js frontend with Vite (dashboard with stats, data tables with search/filter, stock level badges)	Flask API routes, psycopg2 database connection, JSON response formatting, endpoint testing/debugging.		Ability to purchase from inventory

AI Use Disclosure:

Tool(s) used: GitHub CoPilot (Claude Opus 4.6)

Front-end file names only: Inventory.jsx [Opencode - Big Pickle], App.jsx, main.jsx

Team Members: Carlos Mejia, Ashley Prado, Gabriel Garcia, Marcelo Hernandez